

SQL (Structured Query Language)

PL-SQL :-

- PL/SQL stands for Procedural Language / Structured Query Language.
- It is an extension of SQL developed by Oracle Corporation.
- SQL alone can perform only single queries and DML operations, but cannot handle conditions or loops.
- PL/SQL combines the data manipulation power of SQL with the procedural features of programming languages.

1. Procedure :-

- It is just like procedures in other programming languages.
- **Header:** The header contains the name of the procedure and the parameters or variables passed to the procedure.
- **Body:** The body contains a declaration section, execution section and exception section similar to a general PL/SQL block.

Ways to pass parameter

IN

OUT

Syntax

```
DELIMITER //  
  
CREATE PROCEDURE procedure_name  
(parameters...)  
  
BEGIN  
.....statements  
END //  
  
DELIMITER ;
```

Example

```
DELIMITER //  
  
CREATE PROCEDURE insert_user  
(IN p_id int(10),IN p_name varchar(100))  
  
BEGIN  
    insert into user(id,name)values(p_id,p_name);  
END //  
  
DELIMITER ;
```

```
call insert_user(101,'abc');
```

1) Fetch Records :-

```
DELIMITER //  
  
create PROCEDURE fetchrecords()  
  
BEGIN  
    SELECT * from students;  
  
END; //  
  
DELIMITER ;
```

```
call fetchrecords();
```

2) Insert Records

```
DELIMITER //  
  
create PROCEDURE smile(IN id int,IN name varchar(100),IN cid int)  
  
BEGIN  
    insert into students values(id,name,cid);  
  
END; //  
  
DELIMITER ;
```

```
call smile(888,'vijay',999);
```

3) Fetch Single Records

```
DELIMITER //  
create PROCEDURE fetchsingle(IN x varchar(100))  
BEGIN  
    SELECT * from students where sname=x;  
END; //  
DELIMITER ;
```

```
call fetchsingle('Sandip');
```

4) Delete Records

```
DELIMITER //  
create PROCEDURE delbyid(IN k int)  
BEGIN  
    delete from students where sid=k;  
END; //  
DELIMITER ;
```

```
call delbyid(898);
```

2. Trigger :-

- Triggers are stored programs, which are automatically executed or fired when some event occurs.
- Trigger generates some derived column values automatically
- Event logging and storing information on table access
- Imposing security authorizations



```
DELIMITER //  
CREATE TRIGGER trigger_name BEFORE INSERT ON table_name  
BEGIN  
    .....statements  
END //  
DELIMITER ;
```

Example

```
DELIMITER //  
CREATE TRIGGER before_insert_money BEFORE INSERT ON money  
FOR EACH  
ROW  
BEGIN  
    UPDATE employee SET salary='Credited';  
END //  
DELIMITER ;
```

```
insert into money values('100');
```

1) After insert

```
create table patient(pid int primary key,pname varchar(100),pdis varchar(100))
```

```
create table statusinfo(feedback varchar(100)) 'Appotment booked'
```

```
DELIMITER //
```

```
create TRIGGER abc after insert on patient
```

```
FOR EACH
```

```
ROW
```

```
BEGIN
```

```
        insert into statusinfo values('Appoitment Booked');
```

```
END //
```

```
DELIMITER ;
```

```
Select * from statusinfo;
```

```
'Appotment booked
```